United States Patent	12397306
Kind Code	B2
Date of Patent	August 26, 2025
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Spray gun converter

Abstract

Various embodiments of a spray gun converter (**100**) and a spray gun system that includes such converter (**100**) are disclosed. The spray gun converter (**100**) includes a body (**102**) having a first end (**104**) configured for insertion into an air passage of a spray gun (**11**), a second end (**106**) having a converter fitting (**108**) configured for connection to a reservoir (**30**), and a reservoir air passage (**110**) extending through the body (**102**) from an inlet end (**112**) at the first end to an outlet end (**114**) on the converter fitting (**108**) at the second end. The spray gun converter (**100**) also includes a reservoir air control valve (**116**) that opens and closes the reservoir air passage (**110**).

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Appl. No.:	**17/914710**
Filed (or PCT Filed):	**March 23, 2021**
PCT No.:	**PCT/IB2021/052396**
PCT Pub. No.:	**WO2021/191791**
PCT Pub. Date:	September 30, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230145062 A1	May. 11, 2023

Related U.S. Application Data

Publication Classification

Int. Cl.: **B05B7/24** (20060101)

U.S. Cl.:

CPC **B05B7/2416** (20130101); **B05B7/2408** (20130101); **B05B7/2429** (20130101); **B05B7/2478** (20130101);

Field of Classification Search

CPC: B05B (7/2416); B05B (7/2408); B05B (7/2429); B05B (7/2478)

USPC: 239/533.1

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a national stage filing under 35 U.S.C. 371 of PCT/IB2021/052396, filed Mar. 23, 2021, which claims the benefit of U.S. Application No. 63/000,599, filed Mar. 27, 2020, the disclosure of which is incorporated by reference in its/their entirety herein.

FIELD OF INVENTION

(1) The present invention relates to devices and methods for converting a standard gravity-fed spray gun to a pressure-assisted, gravity-fed spray gun. Spray guns are used in a variety of industries to apply liquid coatings to substrates.

BACKGROUND

(2) Spray guns are used, for example, in vehicle repair body shops to apply liquid coating media such as primer, paint and/or clearcoat to vehicle parts. Typically, the spray gun is made of solid metal or plastic and includes a platform and spray head assembly. The spray head assembly includes a nozzle for dispensing the liquid, a center air outlet that provides air to atomize the liquid as it exits the nozzle, and a fan air outlet that provides air to shape the atomized liquid into the desired spray pattern. The spray gun contains a series of internal passages that distribute air from an air supply manifold in the platform to the center air outlet and fan air outlet in the spray head assembly.

(3) Liquid is typically gravity fed to the spray gun by a reservoir that is in fluid contact with the spray head assembly. However, if the liquid becomes too viscous, the spray gun cannot operate as intended, thus preventing delivery of the liquid from the nozzle. In such cases, a pressure-assisted, gravity-fed spray gun (also referred to herein as a “pressure-assisted spray gun”) may be used to provide pressurized air to the reservoir to facilitate fluid flow to the spray gun nozzle.

(4) Traditionally, gravity-fed and pressure-assisted applications required different spray guns. Attempts to provide a single spray gun for both applications have met with limited success. For example, a gravity-fed spray gun has been converted to a pressure-assisted spray gun by adding a secondary regulator below the handle of the spray gun to divert a portion of the air supply to the reservoir. However, the regulator adds bulk to the spray gun, making it less adaptable for applications in tight spaces. In another example, a pressure barb or bolt can be inserted over the air inlet of a pressure-assisted reservoir, thereby causing a pressure-assisted spray gun to perform like a gravity-fed spray gun.

SUMMARY

(5) The present disclosure describes various embodiments of a spray gun converter that can convert a gravity-fed spray gun to a pressure-assisted spray gun by diverting a portion of the air supply to a liquid containing reservoir for pressure-assisted liquid delivery. The spray gun converter can be inserted into a manifold of a platform of the gravity-fed spray gun that feeds air to a nozzle center air outlet and a nozzle fan air outlet of a spray head assembly that is connected to the platform. In at least one embodiment, the spray gun converter is not inserted into the air supply line to the gun (or a separate assembly adjacent to the spray gun body such as a regulator) but rather in the gun body itself. The converter adds little to no additional bulk to the spray gun since the majority of the

converter resides in the platform of the spray gun and, in some cases, replaces a fan control air regulator. Additionally, by placing the converter in the manifold of the platform, the diversion of air to the reservoir has little to no effect on air flow through the nozzle center air and nozzle fan air outlets of the spray head assembly. The spray gun converter allows the user to adjust the amount of air that is diverted to the reservoir and can be turned completely off so that the spray gun operates solely in gravity-feed mode with the converter present. Additionally, or alternatively, the spray gun converter is reversibly attached to the spray gun platform so that it can be quickly and easily inserted or removed, depending upon the desired application.

(6) In one embodiment, the present disclosure provides a spray gun converter that includes a body. The body includes a first end configured for insertion into an air passage of a gravity-fed spray gun, a second end having a converter fitting configured for connection to a reservoir, and a reservoir air passage extending through the body from an inlet end at the first end to an outlet end on the converter fitting at the second end. The spray gun converter also includes a reservoir air control valve that opens and closes the reservoir air passage.

(7) In another embodiment, the present disclosure provides a spray gun that includes a platform having a manifold. The manifold includes a chamber, a manifold air inlet fluidly connected to the chamber, and a manifold center air outlet and a manifold fan air outlet each fluidly connected to the chamber. The manifold further includes an access fluidly connected to the chamber. The spray gun also includes a spray head assembly connected to the platform. The spray head assembly includes a nozzle fluid passage having a nozzle fluid inlet and a nozzle fluid outlet, a nozzle center air passage having a nozzle center air inlet and a nozzle center air outlet, and a nozzle fan air passage having a nozzle fan air inlet and a nozzle fan air outlet. The nozzle center air inlet is fluidly connected to the manifold center air outlet, and the nozzle fan air inlet is fluidly connected to the manifold fan air outlet. The spray gun further includes a spray gun converter that has a body that includes a first end configured for insertion into the access of the manifold of the platform, and a second end having a converter fitting configured for connection to a reservoir. The converter further includes a reservoir air passage extending through the body from an inlet end at the first end to an outlet end on the converter fitting at the second end, where the reservoir air passage is configured for fluid connection to the chamber of the manifold of the platform. The spray gun converter also includes a reservoir air control valve that opens and closes the reservoir air passage.

(8) In a further embodiment, the present disclosure provides a spray gun system that includes a platform having a manifold. The manifold includes a chamber, a manifold air inlet fluidly connected to the chamber, and a manifold center air outlet and a manifold fan air outlet each fluidly connected to the chamber. The manifold further includes an access fluidly connected to the chamber. The spray gun also includes a spray head assembly connected to the platform. The spray head assembly includes a nozzle fluid passage having a nozzle fluid inlet and a nozzle fluid outlet, a nozzle center air passage having a nozzle center air inlet and a nozzle center air outlet, and a nozzle fan air passage having a nozzle fan air inlet and a nozzle fan air outlet. The nozzle center air inlet is fluidly connected to the manifold center air outlet, and the nozzle fan air inlet is fluidly connected to the manifold fan air outlet. The spray gun further includes a spray gun converter that has a body that includes a first end configured for insertion into the access of the manifold of the platform, and a second end having a converter fitting. The converter further includes a reservoir air passage extending through the body from an inlet end at the first end to an outlet end on the converter fitting at the second end, where the reservoir air passage is configured for fluid connection to the chamber of the manifold of the platform. The spray gun converter also includes a reservoir air control valve that opens and closes the reservoir air passage. The spray gun system also includes a reservoir having a reservoir fitting and a fluid outlet that supplies fluid to the nozzle fluid inlet in the spray head assembly, and a reservoir connector attached at one end to the reservoir fitting of the reservoir and at the opposite end to the converter fitting of the spray gun converter.

(9) In a further embodiment, the present disclosure provides a method for converting a gravity-fed

spray gun to a pressure-assisted spray gun. The method includes inserting a spray gun converter into a manifold of the gravity-fed spray gun. The spray gun converter includes a body having a first end configured for insertion into the manifold of the gravity-fed spray gun, a second end having a converter fitting configured for connection to a reservoir, and a reservoir air passage extending through the body from an inlet end at the first end to an outlet end on the converter fitting at the second end. The spray gun converter also includes a reservoir air control valve that opens and closes the reservoir air passage. The method further includes fluidly connecting the converter fitting to the reservoir, and adjusting the reservoir air control valve so that the reservoir air passage is at least partially open.

(10) The term "comprises" and variations thereof do not have a limiting meaning where these terms appear in the description and claims. Such terms will be understood to imply the inclusion of a stated step or element or group of steps or elements but not the exclusion of any other step or element or group of steps or elements. By "consisting of" is meant including, and limited to, whatever follows the phrase "consisting of." Thus, the phrase "consisting of" indicates that the listed elements are required or mandatory, and that no other elements may be present. By "consisting essentially of" is meant including any elements listed after the phrase, and limited to other elements that do not interfere with or contribute to the activity or action specified in the disclosure for the listed elements. Thus, the phrase "consisting essentially of" indicates that the listed elements are required or mandatory, but that other elements are optional and may or may not be present depending upon whether or not they materially affect the activity or action of the listed elements.

(11) In this application, terms such as "a," "an," and "the" are not intended to refer to only a singular entity, but include the general class of which a specific example may be used for illustration. The terms "a," "an," and "the" are used interchangeably with the phrases "at least one" and "one or more."

(12) The phrases "at least one of" and "comprises at least one of" followed by a list refers to any one of the items in the list and any combination of two or more items in the list.

(13) The term "or" is generally employed in its usual sense including "and/or" unless the content clearly dictates otherwise.

(14) The term "and/or" means one or all of the listed elements or a combination of any two or more of the listed elements.

(15) Also herein, all numbers are assumed to be modified by the term "about" and in certain embodiments, by the term "exactly." As used herein in connection with a measured quantity, the term "about" refers to that variation in the measured quantity as would be expected by the skilled artisan making the measurement and exercising a level of care commensurate with the objective of the measurement and the precision of the measuring equipment used. Herein, "up to" a number (e.g., up to 50) includes the number (e.g., 50).

(16) Also herein, the recitations of numerical ranges by endpoints include all numbers subsumed within that range as well as the endpoints (e.g., 1 to 5 includes 1, 1.5, 2, 2.75, 3, 3.80, 4, 5, etc.).

(17) Reference throughout this specification to "some embodiments" means that a particular feature, configuration, composition, or characteristic described in connection with the embodiment is included in at least one embodiment of the disclosure. Thus, the appearances of such phrases in various places throughout this specification are not necessarily referring to the same embodiment of the disclosure. Furthermore, the particular features, configurations, compositions, or characteristics may be combined in any suitable manner in one or more embodiments.

(18) The above summary of the present disclosure is not intended to describe each disclosed embodiment or every implementation of the present disclosure. The description that follows more particularly exemplifies illustrative embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a schematic side view of one embodiment of a spray gun system that includes a spray gun, a spray gun converter, and a reservoir.

(2) FIG. **2** is a schematic perspective view of the spray gun system of FIG. **1**.

(3) FIG. **3** is a schematic cross-sectional view of an air supply manifold of the spray gun of FIG. **1**.

(4) FIG. **4** is a schematic cross-sectional view of a frame of the spray gun of FIG. **1**.

(5) FIG. **5** is a schematic cross-sectional view of a spray head assembly of the spray gun of FIG. **1**.

(6) FIG. **6** is a schematic cross-sectional view of the spray gun of FIG. **1**.

(7) FIG. **7** is a schematic perspective transparent view of the spray gun converter of FIG. **1**.

(8) FIG. **8** is a schematic cross-sectional view of the spray gun converter of FIG. **1** with a reservoir air passage of a body of the converter closed by a reservoir air control valve of the converter.

(9) FIG. **9** is a schematic cross-sectional view of the spray gun converter of FIG. **1** with the reservoir air passage of the body of the converter opened by the reservoir air control valve of the converter.

(10) FIG. **10** is a schematic exploded view of the reservoir of the spray gun system of FIG. **1**.

(11) FIG. **11** is a flow chart of a method for converting a gravity-fed spray gun to a pressure-assisted spray gun.

(12) Unless otherwise indicated, all figures and drawings in this document are not to scale and are chosen for the purpose of illustrating different embodiments of the invention. In particular, the dimensions of the various components are depicted in illustrative terms only, and no relationship between the dimensions of the various components should be inferred from the drawings, unless so indicated.

DETAILED DESCRIPTION

(13) In the following description of illustrative embodiments, reference is made to the accompanying figures of the drawing that form a part hereof, and in which are shown, by way of illustration, specific embodiments. It is to be understood that other embodiments may be utilized and structural changes may be made without departing from the scope of the present invention.

(14) The present disclosure describes various embodiments of a spray gun converter that can convert a gravity-fed spray gun to a pressure-assisted spray gun by diverting a portion of the air supply to a liquid containing reservoir for pressure-assisted liquid delivery. The spray gun converter can be inserted into a manifold of a platform of the gravity-fed spray gun that feeds air to a nozzle center air outlet and a nozzle fan air outlet of a spray head assembly that is connected to the platform. The converter adds little to no additional bulk to the spray gun since the majority of the converter resides in the platform of the spray gun and, in some cases, replaces a fan control air valve. Additionally, by placing the converter in the manifold of the platform, the diversion of air to the reservoir has little to no effect on air flow through the nozzle center air and nozzle fan air outlets of the spray head assembly. The spray gun converter allows the user to adjust the amount of air that is diverted to the reservoir and can be turned completely off so that the spray gun operates solely in gravity-feed mode with the converter present. Additionally, or alternatively, the spray gun converter is reversibly attached to the spray gun platform so that it can be quickly and easily inserted or removed, depending upon the desired application.

(15) The converter can include a reservoir air control valve that is adapted to direct a portion of pressurized air into one or more separate passages. One such passage can be adapted to direct pressurized air to a reservoir. In one or more embodiments, when the valve is in an open position, pressurized air can be directed to the reservoir to pressurize fluid contained therein above atmospheric pressure. Further, when the valve is in a closed position, pressurized air is not directed to the reservoir such that the spray gun operates in a more standard manner with a standard

reservoir.

(16) FIGS. **1-10** are various views of one embodiment of a spray gun system **10**. The spray gun system **10** includes a spray gun **11** having a platform **12** and a spray head assembly **14** connected to the platform. As can be seen in FIG. **5**, the spray head assembly **14** includes a nozzle fluid passage **84** that includes a nozzle fluid inlet **86** and nozzle fluid outlet **16**. The assembly **14** further includes a nozzle center air passage **89** that includes a nozzle center air inlet **81** and a nozzle center air outlet **18**. Further, the assembly **14** includes a nozzle fan air passage **92** that has a nozzle fan air inlet **80** and one or more nozzle fan air outlets **20**.

(17) The platform **12** includes a manifold **22** (FIGS. **3-4**). The manifold **22** includes a chamber **25**, a manifold air inlet **24** fluidly connected to the chamber, and a manifold center air outlet **26** and a manifold fan air outlet **28** each fluidly connected to the chamber. The manifold **22** also includes an access **61** and a second access **62** each fluidly connected to the chamber **25**.

(18) The spray gun **11** also includes a spray gun converter **100** that has a body **102** (FIGS. **7-9**) having a first end **104** configured for insertion into the manifold **22** of the spray gun **11**, a second end **106** having a converter fitting **108** configured for connection to a reservoir **30** (FIGS. **1-2**), and a reservoir air passage **110** extending through the body from an inlet end **112** at the first end to an outlet end **114** on the converter fitting at the second end. The spray gun converter **100** also includes a reservoir air control valve **116** that opens and closes the reservoir air passage **110**.

(19) The spray gun **11** can include a variety of components including the platform **12** and the spray head assembly **14** that can be connected to the platform at a barrel interface **50** (FIG. **4**). The spray gun **11** can include any suitable platform, e.g., one or more embodiments of a platform described in co-owned U.S. Pat. No. 8,590,809 to Escoto, Jr. et al., entitled LIQUID SPRAY GUN, SPRAY GUN PLATFORM, AND SPRAY HEAD ASSEMBLY. In one or more embodiments, the spray head assembly **14** can be releasably connected to the platform **12** and provide features that control movement of both a liquid to be sprayed and air used to spray the liquid as described herein. In one or more embodiments, the spray head assembly **14** is disposable and can be thrown away after use (although in some instances it may be reused). If disposed after use, cleaning of the spray head assembly **14** can be avoided, and the spray gun **11** can be changed over to dispense another liquid by attaching a different spray head assembly to the platform and the same or a different reservoir **30**.

(20) Connection of the spray head assembly **14** to the barrel interface **50** of the spray gun platform **12** may be achieved using any suitable technique or techniques. For example, connection structures on the spray head assembly **14** may cooperate (e.g., mechanically interlock) with openings **52** at the barrel interface **50** to retain the spray head assembly on the spray gun platform **12**. Many other connection techniques and structures may be used in place of those described herein, e.g., a bayonet type connection that facilitates rapid connection/disconnection of the spray head assembly **14** with a push or push-twist action, clamps, threaded connections, etc.

(21) The platform **12** can include any suitable material or materials, e.g., metals, metal alloys, plastics (e.g., polycarbonates, nylons (e.g., amorphous nylons), polypropylenes, etc.), and others. If plastic materials are used to construct the platform **12**, such plastic materials may include any suitable additives, fillers, etc. Selection of the materials used in the platform **12** can be based at least in part on the compatibility of the selected materials with the materials to be sprayed (e.g., solvent resistance and other characteristics may need to be considered when selecting the materials used to construct the platforms).

(22) The manifold **22** within the assembled platform **12** is made up of a variety of cavities that, taken together, form the passages that deliver air to the spray head assembly **14**. For example, FIG. **3** depicts the cavities/air passages formed in the platform **12** with the surrounding structure of the platform removed for clarity. Among the cavities/passages formed in the platform **12** is the manifold **22**.

(23) The manifold **22** includes the chamber **25**. The chamber **25** can take any suitable shape or

shapes and have any suitable dimensions. The manifold **22** further includes the manifold air inlet **24** that can include a fitting **60** (FIG. **4**) such that the manifold can be connected to an air source (not shown) that supplies air to the manifold at greater than atmospheric pressure. The manifold air inlet **24** can take any suitable shape or shapes and have any suitable dimensions. Although depicted as including one manifold air inlet **24**, the manifold **22** can include any suitable number of inlets that can each be fluidly connected to the chamber **25** to supply air or gas to the manifold, e.g., one, two, three, four, five, or more inlets.

(24) The manifold **22** also includes the manifold center air outlet **26** and the manifold fan air outlet **28**. Each of the manifold center air outlet **26** and the manifold fan air outlet **28** can be fluidly connected to the chamber **25** using any suitable technique or techniques. Further, each of the manifold center air outlet **26** and the manifold fan air outlet **28** can take any suitable shape or shapes and have any suitable dimensions. Although depicted as having two outlets **26**, **28**, the manifold **22** can include any suitable number of outlets that are each fluidly connected to the chamber **25**, e.g., one, two, three, four, five, or more outlets. In general, air flowing into the manifold **22** through one or more manifold air inlets **24** can be distributed to one or more outlets **26**, **28** using any suitable technique or techniques.

(25) The manifold **22** can also include any suitable number of accesses, ports, or openings. For example, as illustrated in FIG. **3**, the manifold **22** includes the access **61** and the second access **62** each fluidly connected to the chamber **25** using any suitable technique or techniques. Further, each of the access and the second access **61**, **62** can take any suitable shape or shapes and have any suitable dimensions. Although depicted as including two accesses **61**, **62**, the platform **12** can include any suitable number of accesses, e.g., one, two, three, four, five, or more accesses. The accesses **61**, **62** can be utilized in any suitable manner to provide access to the chamber **25** or to facilitate control of air flow into and out of the manifold **22**. For example, as is further described herein, the first end **104** of the body **102** of the spray gun converter **100** can be configured for insertion into the access **61** of the manifold **22** of the platform **12**. In one or more embodiments, the first end of **104** of the body **102** of the spray gun converter **100** can be configured for insertion into the second access **62** of the manifold **22** of the platform **12**.

(26) Control over both gas or air flow and fluid flow through the spray gun **11** can be accomplished using any suitable technique or techniques. In one or more embodiments, a trigger **64** (FIG. **1**) that is pivotally engaged to the platform **12**, e.g., by a retaining pin **66** can control at least one of the air flow and fluid flow through the spray gun **11**. The spray gun **11** can also include a needle **68** (FIG. **6**) that extends through the spray head assembly **14** in a manner similar to that described in, e.g., U.S. Pat. No. 7,032,839. The trigger **64** can be biased to the inoperative position such that the needle **68** closes the nozzle fluid outlet **16** in the spray head assembly **14** and also closes an air supply valve **70**. In the depicted embodiment, the biasing force is provided by a coil spring **72**, although other biasing mechanisms may be used.

(27) When the trigger **64** is depressed, needle **68** retracts and opens the air supply valve **70** to allow air to pass from the manifold **22** to the spray head assembly **14**. As the trigger is further depressed, the needle **68** is retracted to a position in which a tapered front end **74** of the needle allows liquid to flow through nozzle fluid outlet **16** in the spray head assembly **14**. Air flow can be further controlled by the spray gun converter **100** as is further described herein.

(28) The manifold fan air outlet **28** (FIG. **3**) is fluidly connected to the chamber **25** and routes air to the nozzle fan air outlets **20** of the spray head assembly **14** using any suitable technique or techniques. Further, the manifold center air outlet **26** (FIG. **3**) is fluidly connected to the chamber **25** and routes air to the nozzle center air outlet **18** of the spray head assembly **14** using any suitable technique or techniques. Control over the air distribution from the chamber **25** of the manifold **22** to the manifold center air outlet **26** and the manifold fan air outlet **28** can be accomplished using any suitable technique or techniques as is further described herein.

(29) Connected to the platform **12** of the spray gun **11** is the spray head assembly **14**. As shown in

FIG. **5**, the assembly **14** further includes a barrel **76** and an air cap **78**. The barrel **76** and the air cap **78** of the spray head assembly **14** can combine to form cavities that deliver the center air and the fan air in a substantially isolated manner through the spray head assembly.
(30) The assembly **14** further includes the nozzle fluid passage **84** having the nozzle fluid inlet **86** and the nozzle fluid outlet **16**. The nozzle fluid passage **84** can take any suitable shape or shapes and have any suitable dimensions. Fluid to be sprayed exits the assembly **14** through the nozzle fluid outlet **16**. Fluid enters the nozzle fluid passage **84** from the nozzle fluid inlet **86** that is fed through a fluid port **88**. The nozzle fluid passage **84** can be sized to receive the needle **68** (FIG. **6**) that is capable of closing the nozzle fluid outlet **16** when advanced in the forward direction (to the left in the views depicted in FIGS. **5** and **6**) and opening the nozzle fluid outlet when retracted in the rearward direction (to the right in FIGS. **5** and **6**).
(31) The assembly **14** also includes the nozzle center air passage **89** having the nozzle center air inlet **81** and the nozzle center air outlet **18**. The nozzle center air passage **89** can take any suitable shape or shapes and have any suitable dimensions. The nozzle center air inlet **81** is fluidly connected to the manifold center air outlet **26** using any suitable technique or techniques.
(32) The nozzle center air outlet **18** of the assembly **14** is formed in the air cap **78** such that it at least partially surrounds the nozzle fluid outlet **16**. The spray head assembly **14** is adapted to direct the center air through the nozzle center air outlet **18** at greater than atmospheric pressure. As air exits the center air outlet it pulls liquid or fluid out of the nozzle fluid outlet **16** and atomizes the liquid stream into droplets of a generally conical stream about an axis **2** extending through the nozzle fluid outlet.
(33) The assembly **14** also includes the nozzle fan air passage **92** having the nozzle fan air inlet **80** and the one or more nozzle fan air outlets **20**. The nozzle fan air passage **92** can take any suitable shape or shapes and have any suitable dimensions. Further, the nozzle fan air inlet **80** is fluidly connected to the manifold fan air outlet **28** using any suitable technique or techniques.
(34) The air cap **78** that is provided as a part of the spray head assembly **14** can be connected to the barrel **76** utilizing any suitable technique or techniques. In one or more embodiments, the air cap **78** is connected to the barrel **76** in a manner that allows for rotation of the air cap about the axis **2** relative to the barrel. In one or more embodiments, rotation of the air cap **78** can be used to change an orientation of a pattern of the atomized spray emitted from the spray head assembly **14** relative to the axis **2**.
(35) The air cap **78** can include one or more air horns **90**, each of which defines at least a portion of the nozzle fan air passage **92**. Fan air delivered into the air horns **90** exits through one or more nozzle fan air outlets **20** on the air horns. The outlets **20** on the horns **90** can be located on opposite sides of the axis **2** such that air flowing through the barrel **76** under greater than atmospheric pressure flows against opposite sides of the atomized stream of fluid flowing from the nozzle. The forces exerted by the fan air can be used to change the shape of the stream of fluid to form a desired spray pattern. The size, shape, orientation, and other features of the fan air control outlets **20** can be adjusted to achieve different fan control characteristics as described, e.g., in U.S. Pat. No. 7,201,336 B2 to Blette and entitled LIQUID SPRAY GUN WITH NON-CIRCULAR HORN AIR OUTLET PASSAGEWAYS AND APERTURES. The fan air outlets **20** can take any suitable shape or shapes, e.g., elliptical, rectilinear, etc.
(36) The spray gun system **10** also includes the reservoir **30**. Any suitable reservoir or container can be utilized with system **10**, e.g., one or more embodiments of the containers described in U.S. Pat. No. 7,410,106 to Escoto, Jr., et al. and entitled PRESSURIZED LIQUID SUPPLY ASSEMBLY. As shown in FIG. **10**, The reservoir **30** disclosed herein includes a container **32** that has a side wall **34**, a bottom end **36**, and a top end **38** having a container opening **40** therein. The reservoir **30** also includes a reservoir inlet **42** within the side wall **34**, and a reservoir fitting **44** extending outward from the side wall. The reservoir fitting **44** is adapted to connect the reservoir **30** to a reservoir connector **46** (FIG. **2**). The reservoir **30** can also include one or more additional features, such as

distribution fins to improve air flow and distribution within the container **32**. See, e.g., U.S. Pat. No. 7,410,106.
(37) The container **32** of the reservoir **30** can be adapted to withstand any suitable air pressure. In one or more embodiments, the container **32** is capable of withstanding a relatively high air pressure, e.g., greater than about 69.0 kPa (10 psi), greater than about 137.9 kPa (20 psi), etc.
(38) The reservoir **30** also includes a lid component **160**. The lid component **160** can include a filter component (not shown) either permanently or temporarily attached to a lower surface of the lid component that faces an interior of the container **32** when the lid component is connected to the container. The lid component **160** can further include one or more components capable of connecting to the fluid port **88** of the spray head assembly **14**, where the one or more components are positioned on an outer surface and at a second end of the lid component. For example, the lid component **160** can include axially-spaced radially outwardly projecting sealing rings **162** along an outer surface of cylindrical portion **164** positioned on boss **166**, and opposed inwardly projecting lips **168** on the distal ends of projecting hook members **170**, which are equally spaced from and on either side of cylindrical portion extending from outer surface **172** of exemplary lid component **160**. These various component features may be used to attach the lid component **160** to the fluid port **88**, e.g., as described in U.S. Pat. No. 6,536,687 to Navis et al. and entitled MIXING CUP ADAPTING ASSEMBLY.
(39) The reservoir **30** can also include a liner **174**. The liner **174** can include at least one liner side wall **178**, a liner bottom end **176**, a liner top end **175** having a liner opening **177** therein, and a liner rim **179** extending along and protruding from the liner top end. In one or more embodiments, the liner **174** is self-supporting and collapsible. In one or more embodiments, the liner **174** has a comparatively rigid base **176** and comparatively thin side walls **178** so that the liner collapses in the longitudinal direction by virtue of the side walls collapsing rather than the base.
(40) The reservoir **30** can also include a shroud component **180** that is adapted to provide support to the lid component **160** by extending over and restricting expansion of the lid component when exposed to high pressure. Like the above-described lid component **160**, the shroud component **180** can include an injection-molded part formed from a plastic material such as polypropylene or polyamide. In one or more embodiments, shroud component **180** can be transparent to enable viewing of the lid component **160** and the contents within the liquid supply assembly.
(41) The reservoir **30** can further include a collar **182** that has a top end **183** having a collar opening **185** therein, a bottom end **188**, and at least one collar side wall extending between the top end and the bottom end, a collar rim **190** extending along the top end and protruding into the collar opening, and a set of threads **184** extending along the at least one collar side wall, where the set of threads is capable of engaging with a set of threads **186** of the container **32**.
(42) The reservoir **30** can further include one or more additional, optional components. Suitable optional components include, but are not limited to, a filter element that can be permanently or temporarily attached to the lid component, a gasket that can be positioned between the lid component and the liner (or liner component of the container), an indicating sheet having indicia thereon to assist a user when introducing one or more liquids into the collapsible liner, and an adapter for connecting the lid component to a spraying device positioned between the lid component and the spraying device.
(43) The spray gun **11** also includes the spray gun converter **100**. As shown in FIGS. **7**-**9**, the converter **100** includes the body **102** having the first end **104** that is configured for insertion into the manifold **22** of the platform **12**. The body **102** further includes the second end **106** that has a converter fitting **108** that is configured for connection to the reservoir **30**. Further, the body **102** includes the reservoir air passage **110** that extends through the body from the inlet end **112** at the first end **104** to the outlet end **114** on the converter fitting **108** at the second end **106**. The converter **100** also includes the reservoir air control valve **116** that opens (FIG. **9**) and closes (FIG. **8**) the reservoir air passage **110**. The body **102** can include any suitable material or materials, e.g., at least

one of a metallic, polymeric, or other inorganic material. The body **102** can be manufactured utilizing any suitable technique or techniques, e.g., casting, molding, machining, 3D printing, or any other additive or subtractive manufacturing techniques.

(44) The body **102** can take any suitable shape or shapes and have any suitable dimensions. In one or more embodiments, the body **102** is configured such that at least a portion of the body can be inserted into the manifold **22** of the spray gun platform **12**. For example, the first end **104** of the body **102** can be configured to be inserted into the access **61** of the manifold **22**. In one or more embodiments, the first end **104** of the body **102** can be configured to be inserted into the second access **62** of the manifold **22**. In one or more embodiments, the converter **100** can include a gasket **118** disposed adjacent to the first end **104** of the body **102**. As used herein, the term “adjacent to the first end” means that an element or component is disposed closer to the first end **104** of the body **102** of the converter **100** than to the second end **106** of the body. The gasket **118** can surround at least a portion of the first end **104** of the body **102**. The gasket **118** can include any suitable material or materials and have any suitable dimensions such that the gasket provides an air seal between the body **102** of the converter **100** and the manifold **22**. Although depicted as including two gaskets **118**, the converter **100** can include any suitable number of gaskets, e.g., one, two, three, four, five, or more gaskets. Further, in one or more embodiments, the body **102** can include a slot **120** that is adapted to receive at least a portion of the gasket **118** such that the gasket remains connected to the body **102**.

(45) The second end **106** of the body **102** includes the converter fitting **108**. The converter fitting **108** can be disposed in any suitable location relative to the first end **104** of the body **102**. In one or more embodiments, the converter fitting **108** can be disposed at the second end **106** of the body **102**. In one or more embodiments, the converter fitting **108** is disposed adjacent to the second end **106** of the body **102**. As used herein, the term “adjacent to the second end” means that an element or component is disposed closer to the second end **106** of the body **102** than to the first end **104**.

(46) The converter fitting **108** can take any suitable shape or shapes and have any suitable dimensions. In one or more embodiments, the converter fitting **108** is configured to be connected to the reservoir connector **46** of the reservoir **30** using any suitable technique or techniques. In one or more embodiments, the reservoir connector **46** can be fitted over the converter fitting **108** such that at least a portion of the fitting is disposed within the connector. Although not shown, in one or more embodiments, the converter fitting **108** can include one or more rings or lips disposed on an outer surface of the fitting to retain the reservoir connector **46** when the connector is disposed over the fitting.

(47) Extending through the body **102** of the converter **100** is the reservoir air passage **110**. The passage is configured for fluid connection to the chamber **25** of the manifold **22** of the platform **12**. The passage **110** can take any suitable shape or shapes and have any suitable dimensions. The passage **110** extends from the inlet end **112** of the reservoir air passage **110** at the first end **104** of the body **102** to the outlet end **114** of the reservoir air passage **110** on the converter fitting **108** at the second end **106**. The passage **110** includes a first portion **122** that extends from the inlet end **112** and a second portion **124** that extends from the outlet end **114**. The first and second portions **122**, **124** form an intersection **126** in the body **102** of the spray gun converter **100**. The first portion **122** of the reservoir air passage **110** can form any suitable angle **4** with the second portion **124** as shown in FIG. **8**. In one or more embodiments, the first and second portions **122**, **124** are at substantially right angles at the intersection **126**. As used herein, the term “substantially right angle” means that the angle **4** formed by the first and second portions **122**, **124** of the passage **110** at the intersection **126** is at least 80 degrees and no greater than 110 degrees.

(48) In one or more embodiments, the second portion **124** of the reservoir air passage **110** extends through the body **102** from the outlet end **114** to an opening **128** in the body **102** opposite the outlet end **114**. In one or more embodiments, the second portion **124** linearly extends from the outlet end **114** to the opening **128**.

(49) The converter **100** also includes the reservoir air control valve **116**. The valve **116** is adapted to open and close the reservoir air passage **110**. The valve **116** can include any suitable valve or valves. In one or more embodiments, the valve **116** is a threaded needle valve that includes a reservoir air control knob **132**.
(50) As shown in FIGS. **7-9**, the valve **116** can include one or more threads **134** that are adapted to engage one or more threads **138** disposed in the second portion **124** of the reservoir air passage **110**. The valve **116** can take any suitable shape or shapes and have any suitable dimensions. In one or more embodiments, the valve **116** can include one or more gaskets **140**. The gasket **140** can provide an air seal between the valve **116** and an inner wall of the second portion **124** of the reservoir air passage **110**.
(51) The valve **116** can be located at the intersection **126** of the first portion **122** and the second portion **124** of the reservoir air passage **110**. In one or more embodiments, the valve **116** is threaded into the opening **128** of the second portion **124**.
(52) As mentioned herein, the reservoir air control valve **116** opens and closes the reservoir air passage **110** using any suitable technique or techniques. In one or more embodiments, the valve **116** can be adjusted to close, partially open, or completely open the reservoir air passage **110**. When in the closed position as shown in FIG. **8**, an end **142** of the valve **116** can extend into the second portion **124** of the reservoir air passage **110** and prevent air from the first portion **110** of the reservoir air passage **110** that has entered the passage through the inlet end **112** from passing into the second portion and through the outlet end **114** of the passage. In one or more embodiments, the end **142** of the valve **116** can be shaped to engage a ledge **144** of the second portion **124** of the reservoir air passage **110** to seal the outlet end **114** of the second portion.
(53) Further, as shown in FIG. **9**, the reservoir air control valve **116** can open the reservoir air passage **110** using any suitable technique or techniques. In one or more embodiments, the valve **116** can be rotated such that the end **142** of the valve is withdrawn from the ledge **144** of the second portion **124**, thereby allowing air to flow through the intersection **126** from the first portion **122** and into the second portion. Such air then flows through the outlet end **114** of the second portion **124** and into the reservoir connector **46**, where it is directed into the container **32** of the reservoir **30**. Such air can pressurize the container **32** and provide pressurized fluid to the spray head assembly **14** of the spray gun **11** through the fluid port **88**.
(54) Although the spray gun **11** is illustrated as having the reservoir **30** disposed above the platform **12** when the spray gun is oriented in an in-use position, the spray gun can include a spray head assembly that can be adapted such that the reservoir is disposed below the platform. In such embodiments, the converter **100** can be utilized to provide gas or air to the reservoir **30** to provide pressure-assisted flow if the reservoir air control valve **116** is at least partially opened. Further, with the reservoir air control valve **116** closed, the spray gun **11** would operate as a standard bottom feed spray gun.
(55) In one or more embodiments, the spray gun connector **100** can also include a fan air control valve **146**. The fan air control valve **146** can open and close the manifold fan air outlet **28**. The valve **146** can include any suitable valve. In one or more embodiments, the valve **146** can include a threaded spindle **148** that extends through the body **102** of the spray gun converter **100** from the first end **104** to the second end **106**. The valve **146** can also include a valve plug **150** attached to the threaded spindle **148** adjacent to the first end **104** of the body **102** of the converter **100**.
(56) The fan air control valve **146** can control air delivered to the manifold fan air outlet **28** from the chamber **25**. As a result, the fan air control valve **146** can control air flow to the air horns **90** of the spray head assembly **14** to adjust the spray pattern geometry. Further, the fan air control valve **146** can include a fan air control knob **152** attached to the spindle **148** adjacent to the second end **106** of the body **102**. In one or more embodiments, the threaded spindle **148**, the valve plug **150**, and the fan air control knob **152** can be a single component, i.e., manufactured as one part. In one or more embodiments, the threaded spindle **148**, the valve plug **150**, and the fan air control knob

152 can be manufactured separately and connected together utilizing any suitable technique or techniques.

(57) The reservoir air control valve **116** and the fan air control valve **146** can be disposed in any suitable position relative to each other. For example, in one configuration, the reservoir air control valve **116** is positioned to at least partially open the reservoir air passage **110**, and the fan air control valve **146** is positioned to at least partially open the manifold fan air outlet **28**. In such a configuration, air flows through the reservoir air passage **110**, through the outlet end **114**, through the reservoir connector **46**, and into the container **32**. Such air can be utilized to pressurize the fluid disposed within the container **32** and provide such pressurized fluid to the spray head assembly **14**. Further, in this configuration, air also flows through the manifold fan air outlet **28** and out the nozzle fan air outlets **20**.

(58) In an alternative configuration, the reservoir air control valve **116** is positioned to close the reservoir air passage **110**, and the fan air control valve **146** is positioned to at least partially open the manifold fan air outlet **28**. In yet a further embodiment, the reservoir air control valve **116** is positioned to close the reservoir air passage **110**, and the fan air control valve **146** is positioned to close the manifold fan air outlet **28**. When the reservoir air control valve **116** is closed, the spray gun can be made to operate as a standard gravity fed gun by removing the reservoir connector **46** so that the pressure in the reservoir remains at atmospheric pressure during operation of the gun. Failure to remove the reservoir connector **46** will essentially seal the reservoir so that as the fluid is drawn into the gun, the pressure inside the reservoir drops. This drop in pressure will retard fluid flow from the reservoir and the nozzle fluid outlet **16**.

(59) Any suitable technique or techniques can be utilized with the spray gun system **10** to convert a gravity-fed spray gun to a pressure-assisted spray gun **11**. For example, FIG. **11** is a flowchart of one embodiment of a method **200** of converting the spray gun **11** of the spray gun system **10**. Although described in regard to the spray gun **11** and spray gun converter **100** of the spray gun system **10** of FIGS. **1-10**, the method **200** can be utilized with any suitable spray gun and converter.

(60) At **202**, the spray gun converter **100** is inserted into the manifold **22** of the gravity-fed spray gun **11**. In one or more embodiments, the first end **104** of the body **102** of the converter **100** is inserted into the access **61** or the second access **62** of the manifold **22**. In one or more embodiments, a standard control valve can be removed from the platform **12** if present prior to insertion of the converter **100** into the manifold **22**. When use of a pressurized reservoir **30** is desired, such reservoir is connected to the platform **12** using any suitable technique or techniques. The converter fitting **108** is fluidly connected to the reservoir **30** at **204** using any suitable technique or techniques. In one or more embodiments, at least a portion of the converter fitting **108** can be disposed within the reservoir connector **46**. The reservoir air control valve **116** can be adjusted at **206** so that the reservoir air passage **110** is at least partially open. When the reservoir air control valve **116** is at least partially open, air from the manifold **22** can be directed into the reservoir air passage **110** through the inlet end **112**. Such air can then be directed through the outlet end **114** and into the reservoir connector **46**. This air can be utilized to pressurize the fluid disposed within the container **32**. As discussed herein, the pressurized fluid can be directed to the spray head assembly **14** using any suitable technique or techniques to direct a liquid coating from the system **10** and onto a surface. In embodiments where a standard gravity-fed reservoir **30** is desired, the reservoir air control valve **116** can be manipulated such that the reservoir air passage **110** is closed.

(61) In one or more embodiments, the fan air control valve **146** can be adjusted at **208** so that the valve is at least partially open. When the fan air control valve **146** is at least partially open, air from the manifold **22** can flow into the manifold fan air outlet **28** and through the nozzle fan air outlets **20** of the spray head assembly **14**.

(62) All references and publications cited herein are expressly incorporated herein by reference in their entirety into this disclosure, except to the extent they may directly contradict this disclosure. Illustrative embodiments of this disclosure are discussed and reference has been made to possible

variations within the scope of this disclosure. These and other variations and modifications in the disclosure will be apparent to those skilled in the art without departing from the scope of the disclosure, and it should be understood that this disclosure is not limited to the illustrative embodiments set forth herein. Accordingly, the disclosure is to be limited only by the claims provided below.

Claims

1. A spray gun converter comprising: a body comprising: a first end configured for insertion into a manifold of a spray gun, wherein the manifold comprises a manifold fan air outlet; a second end having a converter fitting configured for connection to a reservoir, and a reservoir air passage extending through the body from an inlet end at the first end to an outlet end on the converter fitting at the second end, wherein the inlet end is configured to fluidically couple to the manifold when assembled; a reservoir air control valve that opens and closes the reservoir air passage; and a fan air control valve, comprising: a threaded spindle extending through the body of the spray gun converter from the first end to the second end; a valve plug attached to the threaded spindle proximate the first end of the body. wherein the valve plug is configured to engage with a portion of the manifold fan air outlet; and a fan air control knob attached to the spindle proximate the second end of the body.
2. The spray gun converter of claim 1, further comprising a gasket surrounding at least a portion of the first end of the body.
3. The spray gun converter of claim 1, wherein the reservoir air control valve is a threaded needle valve comprising a reservoir air control knob.
4. The spray gun converter of claim 1, wherein the reservoir air passage comprises a first portion that extends from the inlet end and a second portion that extends from the outlet end, the first and second portions forming an intersection in the body of the spray gun converter.
5. The spray gun converter of claim 4, wherein the first and second portions are at substantially right angles at the intersection.
6. The spray gun converter of claim 4, wherein the reservoir air control valve is located at the intersection of the first portion and second portion of the reservoir air passage.
7. The spray gun converter of claim 4, wherein the second portion of the reservoir air passage extends linearly though the body from the outlet end to an opening in the body opposite the outlet end, wherein the reservoir air control valve is threaded into the opening, and wherein the reservoir air control valve can be adjusted to close, partially open or completely open the reservoir air passage.
8. The spray gun converter of claim 1, wherein the body comprises a slot disposed adjacent to the first end of the body, the slot is adapted to receive at least a portion of a gasket.
9. The spray gun converter of claim 8, wherein the inlet end is positioned closer to the first end than the slot.
10. The spray gun converter of claim 1, wherein the manifold fan air outlet is fluidly connected to air horns on a spray head assembly of the spray gun, and the air horns comprise nozzle fan air outlets that direct air to shape the atomized liquid into a desired spray pattern.
11. A spray gun comprising: a platform comprising a manifold, wherein the manifold comprises: a chamber; a manifold air inlet fluidly connected to the chamber; a manifold center air outlet and a manifold fan air outlet each fluidly connected to the chamber; and an access fluidly connected to the chamber; a spray head assembly connected to the platform and comprising: a nozzle fluid passage comprising a nozzle fluid inlet and a nozzle fluid outlet; a nozzle center air passage comprising a nozzle center air inlet and a nozzle center air outlet; and a nozzle fan air passage comprising a nozzle fan air inlet and a nozzle fan air outlet; wherein the nozzle center air inlet is fluidly connected to the manifold center air outlet, and the nozzle fan air inlet is fluidly connected

to the manifold fan air outlet; and a spray gun converter comprising: a body comprising: a first end configured for insertion into the manifold fan air outlet; a second end having a converter fitting configured for connection to a reservoir, and a reservoir air passage extending through the body from an inlet end at the first end to an outlet end on the converter fitting at the second end, wherein the inlet end is configured to fluidically couple to the manifold fan air outlet when sealed; a reservoir air control valve that opens and closes the reservoir air passage; and a fan air control valve, comprising: a threaded spindle extending through the body of the spray gun converter from the first end to the second end; a valve plug attached to the threaded spindle proximate the first end of the body, wherein the valve plug is configured to engage with a portion of the manifold fan air outlet; and fan air control knob attached to the spindle proximate the second end of the body.

12. The spray gun of claim 11, wherein the reservoir air control valve is a threaded needle valve comprising a reservoir air control knob.

13. The spray gun of claim 11, wherein the intersection is at substantially right angles.

14. The spray gun of claim 11, wherein the reservoir air control valve is located at the intersection of the first portion and second portion of the reservoir air passage.

15. The spray gun of claim 11, wherein the second portion of the reservoir air passage extends linearly though the body from the outlet end to an opening in the body opposite the outlet end, wherein the reservoir air control valve is threaded into the opening, and wherein the reservoir air control valve can be adjusted to close, partially open or completely open the reservoir air passage.

16. The spray gun of claim 11, further comprising a fan air control valve that opens and closes the manifold fan air outlet.

17. The spray gun of claim 16, wherein the reservoir air control valve is positioned to close the reservoir air passage and the fan air control valve is positioned to at least partially open the manifold fan air outlet.

18. The spray gun of claim 16, wherein the reservoir air control valve is positioned to at least partially open the reservoir air passage and the fan air control valve is positioned to close the manifold fan air outlet.

19. The spray gun of claim 16, wherein the reservoir air control valve is positioned to close the reservoir air passage and the fan air control valve is positioned to close the manifold fan air outlet.

20. The spray gun of claim 16, wherein the reservoir air control valve is positioned to at least partially open the reservoir air passage and the fan air control valve is positioned to at least partially open the manifold fan air outlet.

21. A method for converting a gravity-fed spray gun to a pressure- assisted spray gun, the method comprising: inserting a spray gun converter into a manifold of the gravity-fed spray gun, the spray gun converter comprising: a body comprising: a first end configured for insertion into a manifold of a spray gun, wherein the manifold comprises a manifold fan air outlet; a second end having a converter fitting configured for connection to a reservoir, and a reservoir air passage extending through the body from an inlet end at the first end to an outlet end on the converter fitting at the second end, wherein the inlet end is configured to fluidically couple to the manifold when assembled; a reservoir air control valve that opens and closes the reservoir air passage; and a fan air control valve, comprising: a threaded spindle extending through the body of the spray gun converter from the first end to the second end; a valve plug attached to the threaded spindle proximate the first end of the body, wherein the valve plug is configured to engage with a portion of the manifold fan air outlet; and a fan air control knob attached to the spindle proximate the second end of the body.